

PixCard is an AI-powered SaaS platform that automates **credit scoring**, **risk forecasting**, **compliance reporting**, and **customer segmentation** for banks, lenders, and insurers serving SMEs.

It ingests financial documents (e.g., bank statements), extracts structured data using OCR + NLP, and processes it through ML models to generate explainable credit scores, risk insights, and regulatory reports—all accessible through a real-time dashboard.

1. Document Ingestion & Processing

What It Does:

Transforms unstructured financial documents into structured data for analysis.

How:

- Accepts uploads or API ingestion of PDFs (bank statements, credit card reports, etc.)
 - Uses **OCR** (e.g., Tesseract, Google Vision) to extract text.
 - Applies **NLP** (spaCy, FinBERT) to understand, label, and standardize financial terms.
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2. Credit Scoring Engine

What It Does:

Calculates the creditworthiness of individuals or businesses.

How:

- Uses ML models (XGBoost, LightGBM) to compute a **credit score** based on historical behavior, cash flow, credit utilization, and external risk factors.
 - Outputs a **score and rating** (AAA to D).
 - Uses **Explainable AI (SHAP, LIME)** to justify the decision.
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3. ECL Forecasting (IFRS 9 Compliance)

What It Does:

Predicts future **expected credit losses** (ECL) required by regulators.

How:

- Computes **$ECL = PD \times LGD \times EAD$** using ML models.
 - Ingests **macroeconomic indicators** (e.g., inflation, unemployment).
 - Runs **scenario analysis** (baseline, optimistic, pessimistic).
 - Fully aligns with **IFRS 9** compliance standards.
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4. Customer Segmentation & Risk Monitoring

What It Does:

Groups clients into risk-based categories and monitors them continuously.

How:

- Uses clustering algorithms (e.g., KMeans) to create **risk profiles**.
 - Segments by financial behavior, loan compliance, and fraud likelihood.
 - Flags **high-risk clients** in real-time for manual review.
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5. Automated Compliance Reporting

What It Does:

Generates legally required financial and compliance documentation.

How:

- Produces **IFRS 9 reports**, Fair Lending evaluations, GDPR audit logs.
 - Uses **GenAI (GPT models)** to create human-readable summaries.
 - Exports data in regulator-friendly formats (PDF, JSON, CSV).
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6. Real-Time Dashboard & Analytics

What It Does:

Displays a visual overview of credit risk, forecasts, and alerts.

How:

- React-based dashboard showing **portfolio-level and client-level metrics**.
 - Live updates on credit scores, ECL changes, segment shifts.
 - Embedded **interactive graphs** (D3.js/Plotly) and **alert notifications**.
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7. Workflow Automation & Simulation

What It Does:

Automates data workflows and enables credit simulation for users.

How:

- Uses **n8n or Airflow** for orchestrating document-to-decision pipelines.
 - **Chatbot assistant** (Rasa/Dialogflow) simulates loan eligibility, credit score changes, or helps fill in missing data.
 - Scheduled workflows for retraining models, refreshing reports, and sending alerts.
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8. Security & Access Control

What It Does:

Ensures data protection, secure access, and compliance with privacy laws.

How:

- Implements **RBAC/ABAC**, **AES-256 encryption**, **TLS 1.3**.
- Tracks **audit logs** and session histories.
- Supports **SOC 2**, **GDPR**, **PCI DSS**, and other frameworks.

PixCard automates document-to-decision pipelines for credit risk, combining OCR, ML, explainability, and compliance—all in one platform.

Financial Document Ingestion & Processing (Transforms unstructured financial documents into structured data for analysis.)

Credit Scoring Engine (Calculates the creditworthiness of individuals or businesses.)

ECL Forecasting (Predicts future **expected credit losses** (ECL) required by regulators.)

Customer Segmentation & Risk Monitoring (Groups clients into risk-based categories and monitors them continuously.)

Automated Compliance Reporting (Generates legally required financial and compliance documentation.)

Real-Time Dashboard & Analytics (Displays a visual overview of credit risk, forecasts, and alerts.)

Workflow Automation & Simulation (Automates data workflows and enables credit simulation for users.)

Security & Access Control (Ensures data protection, secure access, and compliance with privacy laws.)

KPI	Description	Target
Credit Score Accuracy	Percentage of accurate credit scores compared to manual assessments	>95%
ECL Forecast Precision	Accuracy of ECL predictions against actual losses	>90%
Processing Time	Time to process a credit application from upload to decision	<5 minutes
Compliance Rate	Percentage of processes compliant with IFRS 9, GDPR, etc.	100%
Fraud Detection Rate	Percentage of fraudulent applications identified	>98%
Client Satisfaction	Net Promoter Score (NPS) from client feedback	>70
Data Ingestion Throughput	Number of documents processed per minute	>100
System Uptime	Percentage of time the platform is operational	>99.9%

Potential Datasets:

1. Financial Document Datasets (OCR/NLP Pipeline)

✓ Real Datasets:

Dataset	Description
SROIE Dataset	Scanned receipts with OCR annotations (for text extraction). Link
FUNSD	Form understanding in noisy scanned docs (annotations for OCR+NLP). Link
RVL-CDIP	Over 400,000 document images from real scanned files. Link
DocBank	Over 500K structured financial and scientific PDFs. Link

Synthetic (Make your own):

- Create **bank statements**, **credit card bills**, **balance sheets**, and **income statements** in PDF format.
 - Use templating libraries (e.g. [pdfplumber](#), [fpdf](#), or LaTeX templates) to render them with randomized numbers.
 - Annotate the key fields (name, amount, date, transaction type) for supervised learning.
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2. Credit Scoring Datasets (ML Models)

✓ Real Datasets:

Dataset	Description
German Credit Dataset	Classic credit dataset with good/bad loan labels. Link
GiveMeSomeCredit (Kaggle)	Over 150K applicants with 10-year default risk. Link
LendingClub Loan Data	Open P2P lending loan data including status, amount, FICO. Link
FICO Explainability Challenge	Contains features used in real credit scoring models. Link

Synthetic:

- Generate transaction histories with:
 - Salary inflows
 - Utility/grocery/loan payments

- Late payments, overdrafts
- Add labels for credit risk: Good (0), Bad (1)
- Add features: income, debt ratio, age, # late payments

Use `scikit-learn`'s `make_classification` + domain constraints.

3. ECL Forecasting (IFRS 9 Compliance)

Real Datasets:

Dataset	Description
IMF Macroeconomic Data	Interest rates, inflation, GDP (for ECL scenario models). Link
OECD Economic Outlook	Country-level macroeconomic time series. Link
European Central Bank SSM Data	Supervisory credit risk datasets. Link

Synthetic:

- Simulate LGD (loss given default), EAD (exposure at default), and PD (probability of default) over time.
- Use regression targets like:
 - $ECL = PD * LGD * EAD$
 - Macroeconomic variables as inputs (GDP, inflation, unemployment)
- Time-series models (ARIMA, LSTM) require sequences—synthesize monthly credit behavior.

4. Segmentation / Customer Behavior Datasets

Real Datasets:

Dataset	Description
Bank Marketing Dataset	Real marketing campaign data with customer financial info. Link
Retail Transaction Data (Instacart)	Behavior-based purchase history to simulate habits. Link
Credit Card Fraud (Kaggle)	Card transactions with fraud labels. Link

Synthetic:

- Simulate customer time-series:
 - Spend patterns (e.g. weekly shopping, bills)
 - Loan repayment consistency
 - Volatility in deposits
- Use this for segmentation: KMeans, DBSCAN, etc.

5. Regulatory/Compliance Data

No open data for this—compliance rules are deterministic logic (IFRS9, Fair Lending, GDPR). You'll:

- Encode rules from docs (e.g., IFRS9 stages: 12-month ECL vs lifetime ECL)
- Generate fake reports using your model outputs

Use **GPT-4o** or **templates** for GenAI report generation.

Bonus Tools to Generate Data

Tool	Use
<code>Faker</code> (Python)	Fake names, addresses, emails, company data
<code>scikit-learn</code>	Synthetic classification/regression data
<code>gretel.ai</code>	Synthetic data generation with structure preservation
<code>fpdf</code> , <code>pdfkit</code>	Render fake financial documents as PDFs
GPT-4o	Generate natural-language financial summaries or fake customer histories

Action Plan

Goal	Dataset
Train OCR/NLP pipeline	FUNSD + Synthetic PDFs
Train credit score model	German Credit + LendingClub
Train ECL pipeline	Kaggle + IMF + synthetic LGD/EAD
Test segmentation	Bank Marketing + synthetic behavior
Simulate real docs	Templated PDFs with Faker + random TXs